Dynamic Equilibrium

Pinning down what stops changing at equilibrium, what never stops, and where the equilibrium ratio actually comes from.

Name ____________________ Date ____________________ Period ____________________

Demonstration: andresdbp.github.io/resources/demos/dynamic-equilibrium/

1 Before you open anything

Answer from what you already know. Getting it right now is not the point — having something specific on paper to argue with later is.

PREDICTION

A box holds 400 particles, all of them A, undergoing A $\rightleftharpoons$ B. Each step, any given A has a 0.060 chance of turning into B, and any given B has a 0.020 chance of turning back into A. Run it until the counts of A and B stop changing.

How many of the 400 particles are A at that point? ___________ And in a typical step at that point, how many particles go A → B? ___________

Write one or two sentences saying how you got those two numbers.

____

____

____

2 Watch one box settle

Open the demonstration. It starts on the **Particles** tab with the sliders already at the values in your prediction.

1. Confirm the two sliders read k_f **= 0.060** and k_r **= 0.020**. Under *Restart the box with*, make sure **All A** is selected, then press **Restart**.
2. Press **Pause** as fast as you can afterwards — aim for *Step* under 10 — and fill in the first row.
3. Press **Resume** and let it run to about *Step* 100, where the *A particles* count stops trending in one direction. Pause and fill in the second row.
4. Press **Resume**, wait until *Step* is past 300, and pause again for the third row.

When	A particles	B particles	A → B / step	B → A / step	Measured [B]/[A]
First few steps					
Around step 100					
Past step 300					

a. Between step 100 and step 300, which quantities in that table changed meaningfully and which did not?

b. In the last row, the two conversion counts are close to equal but the A and B counts are nowhere near equal. State, in one sentence, exactly what quantity has stopped changing at equilibrium.

c. Read the *Measured [B]/[A]* box and the $K = k_f / k_r$ box. Write both values. [B]/[A] = __________ K = __________

> The two conversion counts wobble by a few tenths even after the box has settled, and the measured ratio wobbles with them. That is not sloppiness in the model: the box holds only 400 particles, and each one is flipping a coin. A beaker holds around 10^{21} molecules, and the same wobble is far too small to see.

3 Where the ratio comes from

Now change one rate constant at a time and watch what the equilibrium mixture does. Both sliders move in steps of 0.005, so every value below is reachable exactly.

1. Switch to the **Two Rates, One Line** tab. The readouts you need are $K = k_f / k_r$, *[A] at eq.*, *[B] at eq.* and *Rate at eq.*
2. Leave $\boldsymbol{k_r}$ **= 0.020** alone for this whole part.
3. Set k_f to each value in the table and copy down the four readouts. Total concentration is held at 1.00 mol L^{-1} throughout.

k_f	k_r	K	[A] at eq. (M)	[B] at eq. (M)	Rate at eq.
0.020	0.020				
0.040	0.020				
0.060	0.020				
0.080	0.020				
0.100	0.020				

a. For each row, divide [B] at eq. by [A] at eq. Compare the answer to the K in that row. Write what you find in one sentence.

b. Only one row has [A] = [B]. Which one, and what is special about the two rate constants there? Say why every other row must come out unequal.

c. Using your table, write a formula for [A] at eq. in terms of K alone, given that [A] + [B] = 1.00 M. Test it on the K = 4 row.

d. Go back to your prediction in Part 1. Was either number right? If not, name the specific assumption behind it that the table has just contradicted.

4 Same K, different rate constants

A reasonable shortcut says a bigger rate constant pushes the reaction further toward product. Test it against three settings that all share the same K.

1. Stay on the **Two Rates, One Line** tab.
2. Set each pair below, press **Show both at once**, and record the readouts. Also read the number at the right-hand end of the *time (steps)* axis on the rate plot — that is roughly how long the run takes to flatten.

k_f	k_r	K	[A] at eq. (M)	[B] at eq. (M)	Rate at eq.	Right end of time axis
0.030	0.010					
0.060	0.020					
0.090	0.030					

a. Which columns are identical across all three rows, and which change? What does that say about the shortcut in the intro?

b. Two of these mixtures sit at the same place with very different amounts of traffic across the barrier. Explain what a chemist would actually notice in the lab that distinguishes the first row from the third.

c. *Working backwards.* You want a mixture that ends up 80 % B and 20 % A. Keeping k_r = 0.020, what K do you need and what k_f does that require? Work it out first, then set it and check *[B] at eq.*

d. With that setting still loaded, press **Start from pure A**, then **Start from pure B**. Compare *K from pure A* and *K from pure B*. What does the agreement tell you about what K depends on?

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it, check yourself, and if a number came out wrong, write down where. For all of these, $A \rightleftharpoons B$ is a single-step reversible reaction and the total concentration is 1.00 mol L^{-1}. The *Rate at eq.* readout is rounded to four decimal places, so allow for that when you compare.

1. k_f = 0.075 per step and k_r = 0.025 per step. Find K, [A] at equilibrium, [B] at equilibrium, and the rate at equilibrium. Then state how many of 400 particles would convert A → B in a typical step.

2. A mixture settles at [A] = 0.200 M and [B] = 0.800 M. The reverse rate constant is k_r = 0.020 per step. Find K, find k_f, and find the rate at equilibrium two different ways — once from the forward direction and once from the reverse.

3. k_f = 0.010 per step and k_r = 0.040 per step. Without a calculator, say which side the equilibrium favours and why. Then compute K, [A] and [B].

4. Mixture I runs with k_f = 0.030, k_r = 0.010. Mixture II runs with k_f = 0.090, k_r = 0.030. Mixture II is converting molecules three times as fast in both directions. Explain why Mixture II is not "more product-favoured" than Mixture I, and explain why neither one is "more at equilibrium" than the other.

> Everything above rests on A $\rightleftharpoons$ B being one particle in, one particle out, with both directions happening in a single step — which is what makes K = k_f/k_r exactly true. For a reaction such as $N_2 + 3H_2 \rightleftharpoons 2NH_3$ the coefficients become exponents in K and the rate laws are not this simple. What does carry over everywhere: at equilibrium the forward and reverse rates are equal, and the amounts usually are not.

6 On your own

Nothing here is graded. The demonstration does more than this worksheet uses.

- On the *Particles* tab, restart the box with *All B* and then with *Half each*, leaving the sliders alone. Watch the *Measured [B]/[A]* box in each case. Time how many steps it takes each starting mixture to arrive.
- Push both sliders to 0.010, the slowest setting available, and watch the conversions-per-step plot at equilibrium. Then push both to 0.100. In which case does the random wobble look larger compared with the average, and why should the number of particles in the box matter to that?
- Find every slider pair that triggers the K = 1 warning. What do they all have in common, and what is the same about the equilibrium mixture in every one of them?
- On the second tab, look at the four rate curves after pressing *Show both at once*. Two of them lie exactly on top of each other for the entire run. Work out which two, and why that has to happen.